Data Visualization

Unit - 6

* Bar Chart →

* Purpose →
- A Bar chart is a visual representation of categorical data using rectangular bars.
- The primary purpose of a bar chart is to compare quantities across different categories.
- It provides a clear and simple way to see which categories are larger or smaller in comparison to others, making it ideal for showing relative differences.

* Where to use & Example →
- Bar charts are best used when data is categorical or discrete.
- Scenarios → 1) compare values across different groups
2) Highlight differences or rankings
3) Track changes over time.

e.g. → 1) Comparing the no. of students enrolled in different departments
2) Visualizing the monthly revenue of company.

* Key points :

- Structure
  - X-axis show categories.
  - Y-axis show values.
  - Bars can be vertical or horizontal
  - All bars are of equal width, and spacing is used to seperate them.
- Types → Vertical, Horizontal
  Grouped, Stacked

- Charcterstics →
  - Best for the discrete data
  - Easily interpretable for small data
  - colors or patterns are used to differentiate categories.
  - It can also show positive and negative values.

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* Histogram → * Purpose:
- A histogram is a graphical representation used to display the distribution of a dataset.
- It shows how frequently values occur within specific ranges (bins or intervals).
- It helps to identify patterns such as skewness and outliers.

* where to use & e.g →
- When the data is continuos and numerical (Not categorical).
- To analyze distribution.
- To identify patterns, such as frequent values, gaps & outliers.

e.g →
- showing the distribution of student marks in a maths exam
- Analyzing ages of people in a city population dataset.

* key points →

• Structure:
- X-axis: intervals/bins
- Y-axis: Frequency
- Bars touch each other to show continuity
- Height of each bar indicates no. of values in that interval

• Features:
- No gap between bars.
- Each bar represents a range, not a single value.
- Distribution seen like:
  1) Normal (bell shaped)
  2) left-skewed / right skewed
  3) Uniform

* Advantages
- Understand large dataset helpful
- Used for detect outliers

* Limitations:
- Not for categorical data
- There is problem while set the bin size
- Exact data values are not visible.

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Scatter plot

* Purpose: →

- A Scatter plot is a type of graph used to show the relationship (correlation) between two continuous numerical variables.
- Each point on the plot represents an observation in the dataset, with its x and Y coordinates determined by the values of the two variables.
- The main purpose of a scatter plot is to visually examine there is a trend, pattern or correlation between the variables.

* Where to use & example →

- You want to analyze patterns or trends between two numeric variables.
- You want to detect correlations or relationships.
- You have to working with bivariate data (two numeric

e.g. →

- Height vs weight of individuals
- Study hours vs Exam scores

* Key points →

• Structure:

- X-axis → independent variable.
- Y-axis → dependent variable.
- Each point (dot) is pair of values
- No bars or line - only dots.

• Types of Relationships -

1) Positive corr 2) Negative corr
3) No corr 4) Non linear rel^n

• Features →

- Detects outliers (far point from
- Helps in regression analysis
- can show clustering of data.

* Advantages →

- Simple
- identify patterns & corr^n
- Useful in EDA

* Limitations →

- only for the two variables
- Not for categorical data

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Strip plot →

* Purpose →

- A strip plot is a type of plot used to display the distribution of a small dataset across categories.
- It shows individual data points along a single axis, typically the Y-axis or X-axis, and is especially useful when you want to visualize all data points instead of aggregated summaries like in bar & box plot.
- The main purpose is to observe the spread, clustering and potential overlaps of data points in categorical values or groups. for small datasets.

* Where to use & example →

- When you want to compare data points across different categories.
- Useful for small datasets or when you want to see each individual data point.
- often used alongside boxplots or violin plot for detailed vie

e.g. →

- Comparing exam scores of students from diff. departments
- Showing product rating by diff customer groups.

* Key points →

- structure →
- X-axis: Categorical variable
- Y-axis: numerical values
- Each dot represent single observation
- Dots may be jittered to avoid overlap.

* Advantages -

- simple
- too small datasets (boxplot small)
- comparison between two categori

* features →

- Shows exact values for each.
- Maintains categorical grouping.
- Easier to interpret small dataset
- It can reveal gaps & outliers.

* Limitations →

- Not used for large datasets (overlapping)
- Not show statistical theory like median or quartiles (unlike boxplot)

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Violin Plot → * Purpose

- A violin plot is a combination of a boxplot and a density plot that displays the distribution of numerical data across different categories.
- It provides more detailed view of the distribution of the data by showing both the quantiles (box plot) and the probability density of a variable (density plot).
- The main purpose of a violin plot is to visualize the distribution, density and spread of data across categories, enables you to observe the shape of data, detect skewness and identify outliers in a more standard way than box plot.

* When to Use & e.g. →

- When you want to compare distributions across multiple categories
- When you need to see the density of the data, which helps to identify patterns that box plot can't
- To observe skewness & multi-modal distributions that are often hidden in simpler plots.

e.g →

- comparing the test scores of different groups of students (e.g. male vs female)
- Analyzing salary distribution.

* Key points

- structure →
- X-axis: Categorical value
- Y-axis: numerical value
- The plot has box plot inside it and shows mean, IQR & outliers
- The density plot shows the distribution of the data for each category.

* Feature →

- Displays distribution shape.
- Width indicates the data density for that value allowing us to see.

* Limitations:

- Misleadings for small datasets
- Overlapping violins in multi-category
- Take large space.

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* Swarm plot →

* Purpose:

- A swarm plot is a type of data visualization used to display individual data points along a single axis, where each point is differently spaced to avoid overlapping.
- It is a variation of strip plot, but with improved handelling of data points, especially when there is a large number of observation.
- Main Purpose:
  - Show individual data points clearly, makes it easy to observe the distribution, density and patterns.
  - Display small datasets or situations where you want to emphasize every single data point.
  - Avoid overlap of data point, ensuring that each data point is visible, unlike in traditional strip plots where point might overlap.

* Where to use & e.g →

- small to medium-sized datasets where individual points need to be clearly visible.
- when you want to avoid data point overlap, making it ideal for categorical comparisons across groups.
- Useful when working with discrete data or when you need to emphasize the density and distribution of data.

e.g. →

- Exam scores for students across different classes, showing how each individual score is distributed in each class.
- Product ratings for different products where you want to see how each rating is distributed across the product categories.

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* Key points :
  - Structure :
  - X-axis → Categorical values
  - Y-axis → Numerical values
  - Data points are placed along the axis, with adjustments made to prevent overlapping, ensuring all points are visible.

- Features →
  - Non-overlapping points: no overlapping
  - Spacing between points : clear views of data distribution
  - Comparison of distribution across multiple categories.
  - It allows you to detect outliers and observe how dense or spread out the data is within each category.
  - Mainly for small datasets

* Limitations :-
  - Can be ineffective for large datasets
  - May not be ideal for continuous variable, leads to much overlapping.
  - Hard to implement for large datasets.

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* Pairplot →

- A pairplot is used to visually explore the relationships between multiple numerical variables in a dataset by plotting pairwise scatterplots. It is especially for the EDA.
- Purpose:
  - Understand how different features relate to each other
  - Detect correlations, clusters, trends or outliers.
  - Identify relationships between variable across different categories (with hue).
- Each diagonal of plot shows distribution (histogram) of individual variable, while off-diagonal shows scatterplots between variable pairs.

* Where to use & Example:

- During EDA to analyze multiple variables together.
- when working with multivariate data (3 or more features)
- To detect multicollinearity between features.
- To visualize class seperation when a categorical label used

e.g. →

{Height, Weight, Age, income} of people we pair plot for relate features.

* Key points

- Structure
- It's a grid of subplots
- Each cell in the grid is a scatterplot
- The diagonal often shows univariate distribution
- You can use the hue to group data,

• Features →

- shows pairwise relation
- visualize linear or non-linear
- Good for identify best features
- Use in classification.

* Limitations:

- Not ideal large datasets
- only works well with
  - numeric features
- For very high-dimensional data,

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* Joint plot → * Purpose →
- A joint plot is a powerful visualization tool that combines a bivariate scatter plot with univariate distributions of individual variables.
- It's main purpose is to show both the relationship between two variables and the distribution of each variable in a single figure.
- It provides a comprehensive view of how two variables relate, and how each variable behave on its own.

* Where to use & example →
- when you want to explore correlation or pattern between two numeric variables.
- During EDA to check for linear or non-linear relations
- When you need both distributional insight and bivariate relationships.
- When visualizing data spread, density & outliers.

e.g. →
If you have dataset containing Hours studied & exam score you can use Joint plot to
- Display a scatter plot of Hours vs Score
- show histograms of Hours studied and Exam score on top &

* Key points:
• structure:
- Central plot: scatter (default) or regression
- Top Axis: Distribution (histogram) of X-axis variable.
- Right axis :. Distribution (histogram) of y-axis variable.

* Advantages:
- two types of plot in one plot
- Efficient for quick insights

* Disadvantages →
- only works for two variables at a time
- can become cluttered with large

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